

The power of offloading.

What the chat logs show, and
what I changed in my own course.

EEG during essay writing

MIT Media Lab, 2025 · 54 students

Brain only strongest connectivity

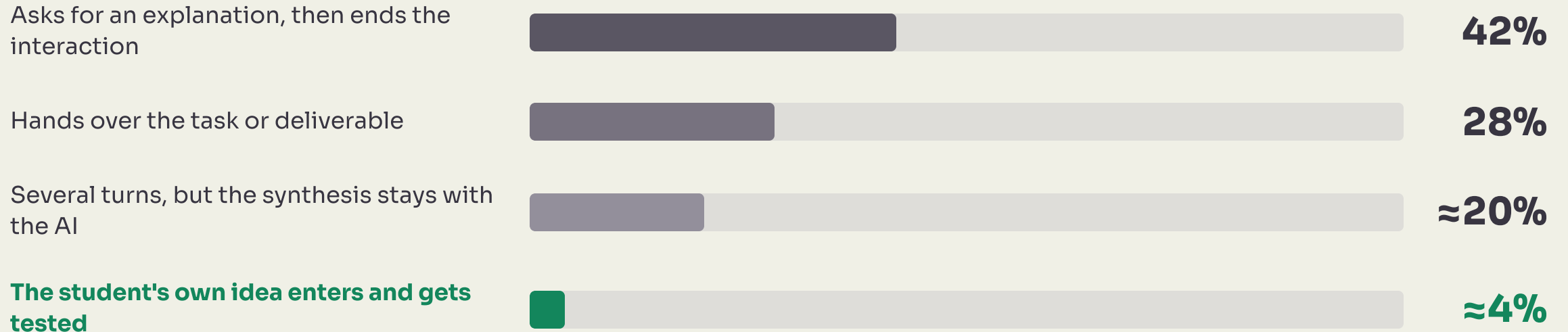
Search engine in between

LLM **weakest**

The LLM group showed weaker brain connectivity.

In search of constructive use

40,000 student messages to an AI tutor



Preliminary, descriptive categorisation

Tamási, *The Power of Outsourcing*, The European Sociologist 55, 2026

Our assignments reward the result

What I saw

Most conversations end as soon as the answer arrives.

Why

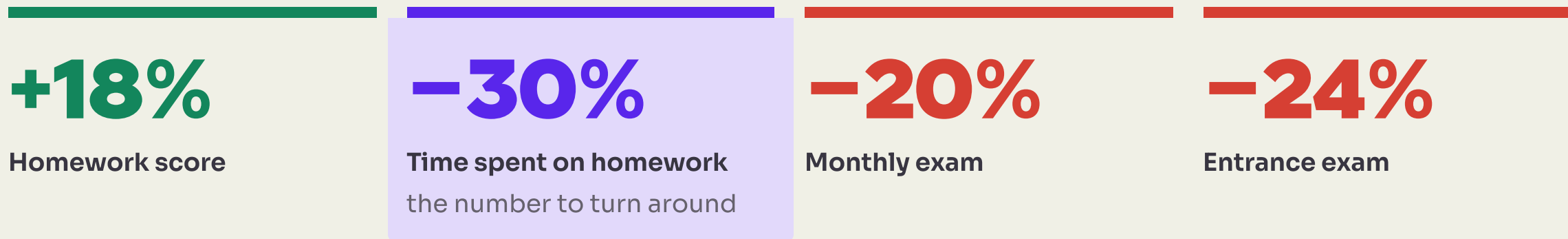
If the result is what counts, the shortest path to it is the rational choice.

So

The finished result proves less and less about the student's thinking.

Better homework, weaker exams

26,811 Chinese secondary school students, 30 months



+18%

Homework score

-30%

Time spent on homework
the number to turn around

-20%

Monthly exam

-24%

Entrance exam

Students who kept spending the time barely
lost ground. **The lever is time on their own work.**

Our attempt: reward the process

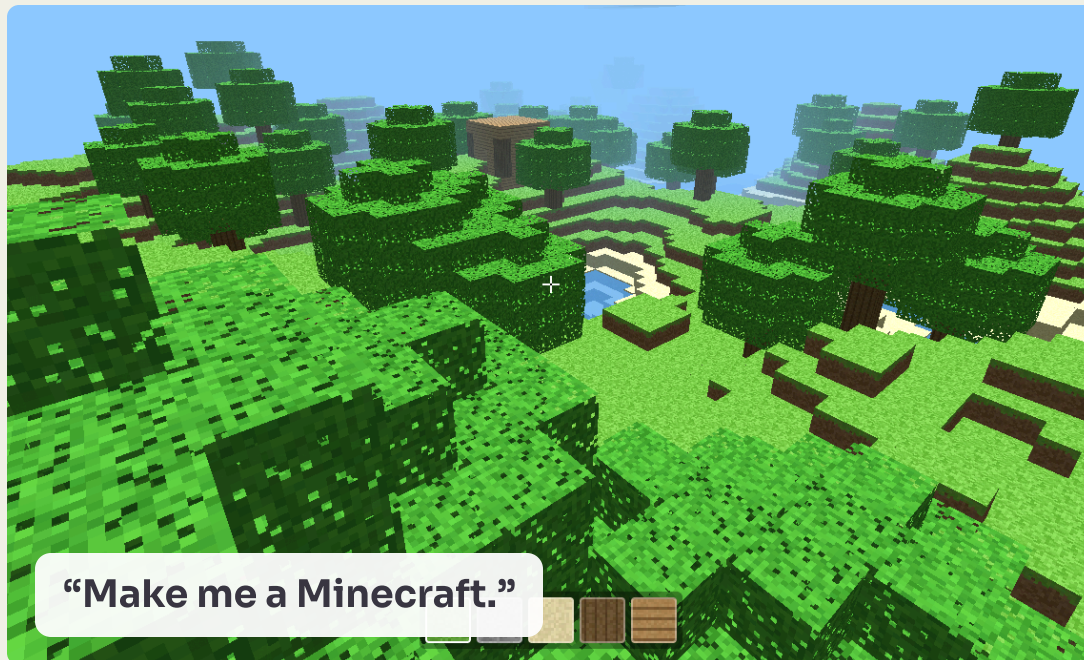
Two courses, 2025/26 - a course design trial, not an impact study

Two programs, both built with AI

Vibe coding raises the floor, agentic engineering raises the ceiling.

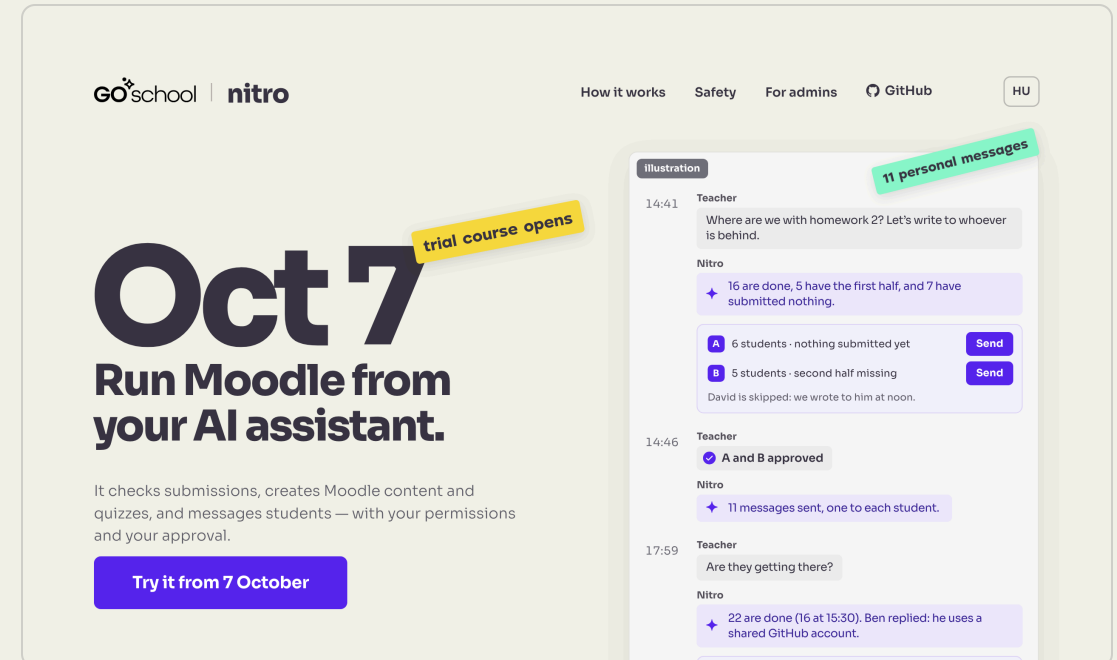
Vibe coding

a playable 3D world from one sentence



Agentic engineering

a Moodle plugin that teachers rely on



Own project, own spec, own mistakes

Two courses: 28 students and 16 teachers, 44 own projects

The -30%, turned around.

Goal: more time on their own work, not less.

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- 01 Motivated to work on it
 - 02 Nothing to copy
 - 03 Helping each other is pure gain
 - 04 It's what work will expect
-

Leading them through the process

Weekly steps, so the process shows week by week

Project

A meaningful, well-defined problem

Plan

They revise it when the AI suggests better

Specification

They shape it, not just nod along

Spike

They test the risky points
with a quick experiment

For every word of mine, the AI read 44 and wrote 45

My two courses, September 2026 · 1,031 requests to the AI



15k

words I wrote

655k

words the AI read

681k

words the AI wrote

Time: about 115 hours in sessions, about 40 of them the AI working.

Don't fight offloading. Use the time it frees.

Which of your tasks would make that
time worth the student's own work?



arpadtamasi.github.io/innofair-2026

discussion